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1 Elimination

A fundamental technique for finding the solutions of a system of linear equations is that of *elimination* of variables. Roughly, this technique involves multiplying the equations in the system by numbers and then adding the resulting equations together so that some of the variables drop out, leading to a simpler system of equations.

Example 1.1 (Solving by elimination)**.** To illustrate this technique, consider the homogeneous system

$$\begin{aligned} 2x_1 - x_2 + x_3 &= 0 \\ x_1 + 3x_2 + 4x_3 &= 0. \end{aligned}$$

Multiplying the second equation by (-2) and adding it to the first gives $-7x_2 - 7x_3 = 0$, which implies

$$x_2 = -x_3. \tag{1.1}$$

Now muliplying the first equation by 3 and adding it to the second gives $7x_1 + 7x_3 = 0$, which implies

$$x_1 = -x_3. \tag{1.2}$$

Thus, a solution of this system is obtained by setting $x_3 = t$, where t can take any value, and then solving for x_1 and x_2 in terms of t using (1.1) and (1.2). The solution set can therefore be written as $\{(-t,-t,t) : t \in F\}$, which is said to be written in *parametric form*. We see that there is a solution for each $t \in F$, so this system has an infinite number of solutions.

1. Verify that $(-t,-t,t)$ is indeed a solution of the system of equations in the previous example for any $t \in F$.

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We now begin to formalize the elimination process in order to carry it out in a systematic way and to understand why it works. Consider a general linear system of m equations in n unknowns

$$\begin{aligned} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n &= y_1 \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n &= y_2 \\ \vdots \qquad\quad \vdots \qquad\qquad \vdots \quad \vdots & \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n &= y_m. \end{aligned} \tag{1.3}$$

If we select $c_1, \cdots, c_m \in F$, multiply the jth equation by c_j for $j = 1, \cdots, m$, and add all of the equations together, we obtain a new linear equation of the form

$$\alpha_1 x_1 + \cdots + \alpha_n x_n = \beta \tag{1.4}$$

2. Give explicit formulas for α_k and β in terms of A_{ij}, y_k, c_ℓ.

Definition 1.5 (Linear combination)**.** The equation (1.4) is said to be a *linear combination* of the equations in (1.3).

Proposition 1.6 (Linear combinations of linear equations preserve solutions)**.** *If* $(s_1, \cdots, s_n)$ *is a solution of the linear system* (1.3)*, then it is also a solution of* (1.4)*.*

3. Prove Proposition 1.6

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Proposition 1.6 gives the fundamental idea of the elimination process: given the linear system (1.3), if we have another linear system

$$\begin{aligned} B_{11}x_1 + B_{12}x_2 + \cdots + B_{1n}x_n &= z_1 \\ B_{21}x_1 + B_{22}x_2 + \cdots + B_{2n}x_n &= z_2 \\ \vdots \qquad\quad \vdots \qquad\qquad\quad \vdots \quad \vdots & \\ B_{k1}x_1 + B_{k2}x_2 + \cdots + B_{kn}x_n &= z_k \end{aligned} \tag{1.7}$$

in which each of the k equations is a linear combination of the equations in (1.3), then *every solution of* (1.3) *is a solution of this new system*. Thus, *the elimination process allows us to solve a linear system by forming linear combinations of the given equations to produce an system of equations which has the same solutions, but is easier to solve*.

Of course, it may happen that some solutions of (1.7) are not solutions of (1.3). As a simple example, consider the following two linear systems:

$$x_1 - 2x_2 = -1 \tag{1.8}$$
$$-x_1 + 3x_2 = 3 \tag{1.9}$$

and

$$x_1 - 2x_2 = -1 \tag{1.10}$$
$$-x_1 + 2x_2 = 1 \tag{1.11}$$

4. (a) Use the elimination technique to show that every solution of the first system is a solution of the second, but the second has solutions that are not solutions of the first.

   (b) Show that both equations of the second system can be written as linear combinations of the equations in the first system, but that both equations of the first system cannot be written as linear combinations of the equations in the second system.

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Definition 1.12 (Equivalent linear systems)**.** Two systems of linear equations are said to be *equivalent* if they have the same set of solutions.

We have just seen that two linear systems might fail to be equivalent if the equations in one system cannot be written as linear combinations of the equations in the other system. The following theorem gives a sufficient condition for two linear systems to be equivalent.

Theorem 1.13 (Equivalent linear systems)**.** *Two systems of linear equations are equivalent if each equation in each system is a linear combination of the equations in the other system.*

5. Use Proposition 1.6 to prove Theorem 1.13.

If the elimination process is to be effective in finding the solutions of the system (1.3), then we must see how to form linear combinations of the given equations to produce an equivalent system of equations which is easier to solve. In the next section, we discuss a method to systematically do this.